SGPA: Spectrogram-Guided Phonetic Alignment for Feasible Shapley Value Explanations in Multimodal Large Language Models

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Abstract

End-to-end audio language models process speech without explicit linguistic structure, making it difficult to identify which parts of an utterance drive a given response. Shapley value attribution offers a theoretically grounded remedy, but applying it to audio is intractable under native tokenization: a typical utterance yields over 150 encoder frames, inflating the coalition space by roughly 10^{42} relative to text; individual frames lack standalone meaning; and token boundaries that bisect phonetic transitions introduce masking artifacts. We introduce Spectrogram-Guided Phonetic Alignment (SGPA), a four-stage pipeline that combines Connectionist Temporal Classification forced alignment with spectral boundary refinement to produce acoustically stable, word-aligned audio segments. Controlled diagnostics on LFM2-Audio-1.5B with VoiceBench show that SGPA yields a 43× reduction in model evaluations. Statistical testing confirms that SGPA significantly alters attribution concentration while preserving the global cumulative profile, establishing it as a feasibility-enabling layer for audio explainability.

Index Terms: Audio Processing, Speech Interpretability, Explainable Artificial Intelligence, Multimodal Large Language Models, Shapley Values

1. Introduction

The rapid evolution of Multimodal Large Language Models (MLLMs) has produced systems that process audio end-to-end, bypassing traditional cascaded speech-to-text pipelines [1, 2]. While these architectures achieve strong performance, their internal decision-making remains opaque: when an audio input drives a particular response, there is no principled way to determine which parts of the signal were most influential [3]. Shapley value (SV) attribution [4], popularized in machine learning by Lundberg and Lee [5] and recently extended to token-level LLM explanations by Horovicz and Goldshmidt [6], offers a theoretically grounded framework for this analysis. However, applying SV to audio inputs introduces barriers that do not arise for text [7].

The root cause is native audio tokenization. Unlike text, which discretizes naturally into words or sub-words, audio is processed as a sequence of dense encoder frames [1]. This representation introduces three critical barriers for SV-based audio analysis:

- **Dimensionality explosion.** A standard 3-second utterance produces over 150 encoder frames under native tokenization. Since exact SV computation requires evaluating all 2^n subsets, this inflates the coalition space from 2^{10} (for a 10-word text) to 2^{150} – a factor of $\approx 10^{42}$. For a representative 50-token prompt, exact computation would require $\approx 10^{15}$ model evaluations; at 1 ms per call this exceeds 30 years.[1]
- **Semantic dilution.** Individual 20 ms audio frames lack standalone meaning. Attributions assigned to these opaque units appear as granular noise that cannot be interpreted by human supervisors or linguistic experts [8].
- **Boundary artifacts.** Native token boundaries frequently bisect phonetic transitions such as co-articulations [9, 10]. Masking tokens at these positions introduces audible discontinuities and out-of-distribution perturbations that degrade the fidelity of SV estimates.

This work addresses these challenges by introducing **Spectrogram-Guided Phonetic Alignment (SGPA)**, a four-stage preprocessing pipeline that maps raw audio waveforms to word-aligned segments using Connectionist Temporal Classification (CTC) [11] and local spectral refinement [12]. By replacing architecture-specific encoder frames with semantically grounded word-level segments, SGPA reduces the effective player count by 10–50×, compressing the required model evaluations from ≈2,552 to ≈59 per sample – making SV-based audio attribution computationally feasible on consumer hardware.

Our contributions are threefold: (i) we present SGPA, a model-agnostic alignment layer for audio SV computation; (ii) we provide controlled diagnostics demonstrating that SGPA is not a neutral transformation – it significantly alters attribution concentration (Cohen's d [13] up to -1.37) while preserving the global cumulative SV profile; and (iii) we release an open-source Python package, `mllm-shap`[2], implementing the full pipeline. SGPA is integrated into a broader explainability platform available in the accompanying package [14].

The central contribution is not the individual components – CTC alignment and spectral boundary refinement are established tools – but their composition into a feasibility-enabling layer that makes SV attribution on audio computationally tractable and linguistically interpretable.

Section 2 details the SGPA pipeline. Section 3 presents diagnostic experiments, and Section 4 discusses limitations and future directions.

2. Spectrogram-Guided Phonetic Alignment

In Shapley value attribution, a *player* is an atomic input unit whose contribution to a model's output is being quantified; a

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[1] Derived from $2^{50} \times 10^{-3}\,\text{s} \approx 3.6 \times 10^{10}\,\text{s} \approx 35$ years.

[2] `https://pypi.org/project/mllm-shap/`

coalition S is any subset of players presented to the model, with absent players replaced by a neutral baseline (here, silence). SGPA redefines the player set: instead of opaque encoder frames, each word in the transcript becomes a single player, whose removal is implemented by silencing its corresponding audio segment. This reformulation reduces the player count to a function of the number of words, making the coalition space manageable for SV estimation compared to native tokenization.

SGPA is a four-stage pipeline that converts a raw audio waveform and its transcript into word-aligned audio segments suitable for SV analysis. When a player is excluded from a coalition, its corresponding audio segment is replaced with silence (zero amplitude) [15], preserving the temporal structure of the remaining signal. Figure 1 provides an overview.

2.1. Stage 1: Transcript Decomposition

The input transcript is decomposed into words and characters, recording a character-to-word membership map. This bookkeeping is essential because Stage 2 operates at the character level, while the final SV game requires word-level segments. The membership map allows character-level time boundaries to be merged upward in Stage 4.

Because Stage 2 operates at the character level using a Wav2Vec2 character vocabulary, no word is strictly out-of-vocabulary; however, uncommon phoneme sequences may yield lower-confidence alignment timestamps, which Stage 3 partially compensates for by seeking spectrally stable cut regions within the inter-character gap rather than at a fixed offset from the raw boundary.

2.2. Stage 2: Initial Alignment via CTC

We feed the audio waveform into Wav2Vec2-XLSR-53 [16], a self-supervised speech model whose CTC alignment quality has been validated on forced-alignment tasks [17, 18] and is publicly available; its multilingual pretraining additionally enables future cross-lingual extension. The model produces an emission matrix $\mathbf{E} \in \mathbb{R}^{T \times V}$, where T is the number of time frames and V the vocabulary size. We apply Viterbi decoding [19] to extract the most likely character path:

$$\pi^* = \arg\max_{\pi} P(\pi \mid \mathbf{E}) \tag{1}$$

This yields approximate start and end frame indices for each character, converted to timestamps via the model's effective frame stride. These timestamps serve as coarse anchors but are not guaranteed to coincide with acoustically stable cut points.

Blank tokens dominate CTC emission sequences. In our implementation, blank-labelled frames are excluded from character spans by construction: a character's span runs from the frame where that character first becomes the active non-blank token to the frame where the *next* token transition occurs, regardless of whether that transition leads to another non-blank token or to a blank. Inter-character and inter-word blank regions therefore manifest as *gaps* between adjacent character spans – explicit time intervals with neither endpoint assigned a character label. These gaps are the primary input to Stage 3, which resolves each gap into a single refined boundary timestamp.

2.3. Stage 3: Spectral Boundary Refinement

CTC-derived character boundaries may land in the middle of phonetic transitions. Stage 3 refines each boundary by searching locally for the acoustically quietest cut point, using two signal-level features computed with `librosa` [12]:

- **Short-time energy** $E[n]$: the RMS energy per frame, capturing local amplitude.
- **Spectral flux** $SF[n]$: the frame-to-frame L^2 change in magnitude spectrum, derived from the Short-Time Fourier Transform.

The refined boundary is selected as the point of minimal combined activity:

$$t_{\text{final}} = \arg\min_{t \in W} \left(\alpha\, E[t] + \beta\, SF[t]\right) \tag{2}$$

where $\alpha = 0.8$ and $\beta = 1 - \alpha = 0.2$ were determined through a two-stage grid search over $\alpha \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ using artificial audio from the experimentation dataset (Subsection 3.1). In Stage 1, an automated proxy click-artifact score—based on transient energy and high-frequency power at detected masking boundaries—was computed for all samples to generate a machine-ranked shortlist. In Stage 2, a single annotator manually evaluated a 20-sample subset using a five-point Mean Opinion Score (MOS) scale ($1 =$ severe clicks, $5 =$ no clicks). Details in Appendix A.

Three design choices distinguish the refinement procedure from a naive sliding-window minimum:

Gap-midpoint anchoring. Each inter-character gap $[t_L, t_R]$ – where t_L is the end of the left character's span and t_R is the start of the right character's span – defines both the anchor and the extent of the search window: the window is centred on $(t_L + t_R)/2$ with half-width $\delta = (t_R - t_L)/2 + 40\,\text{ms}$. Anchoring at the gap midpoint rather than at either character's raw boundary ensures symmetric coverage of the blank-dominated silence regardless of gap duration, preventing the search from being confined to one character's acoustic tail when the inter-word pause is long. The $40\,\text{ms}$ padding was chosen heuristically to accommodate transient boundary effects such as stop consonant release bursts and short-lag voice onset times, which typically fall within 0–25 ms [20], providing a comfortable margin above the upper end of that range. The $80\,\text{ms}$ fixed-window fallback was set to encompass the full range of English aspirated stop VOTs (60–100 ms) [20], consistent with the segmental closure durations documented for connected English speech [21]. While these values ensure robust coverage of local acoustic uncertainty, their exact optimality was not determined via grid search and warrants further empirical exploration.

Joint boundary resolution. Each shared boundary between adjacent character spans is resolved *once*. The single refined time is assigned as the end of the left character and the start of the right character simultaneously, and the result is clamped to $[t_L, t_R]$. This joint resolution guarantees that character spans remain monotonically ordered and non-overlapping, which is a precondition for artifact-free segment masking in Stage 4.

Silence-aware fallback. The minimum-cost frame is accepted only if its RMS energy falls below half the window mean – a threshold indicating a genuine inter-word pause. In fast or continuous speech where no such frame exists, the algorithm falls back to the raw gap midpoint rather than forcing a cut inside an active phoneme. Each resulting `AudioSegment` carries a `boundary_refined` flag (set to `False` for fallback cases) that propagates through word-level aggregation,

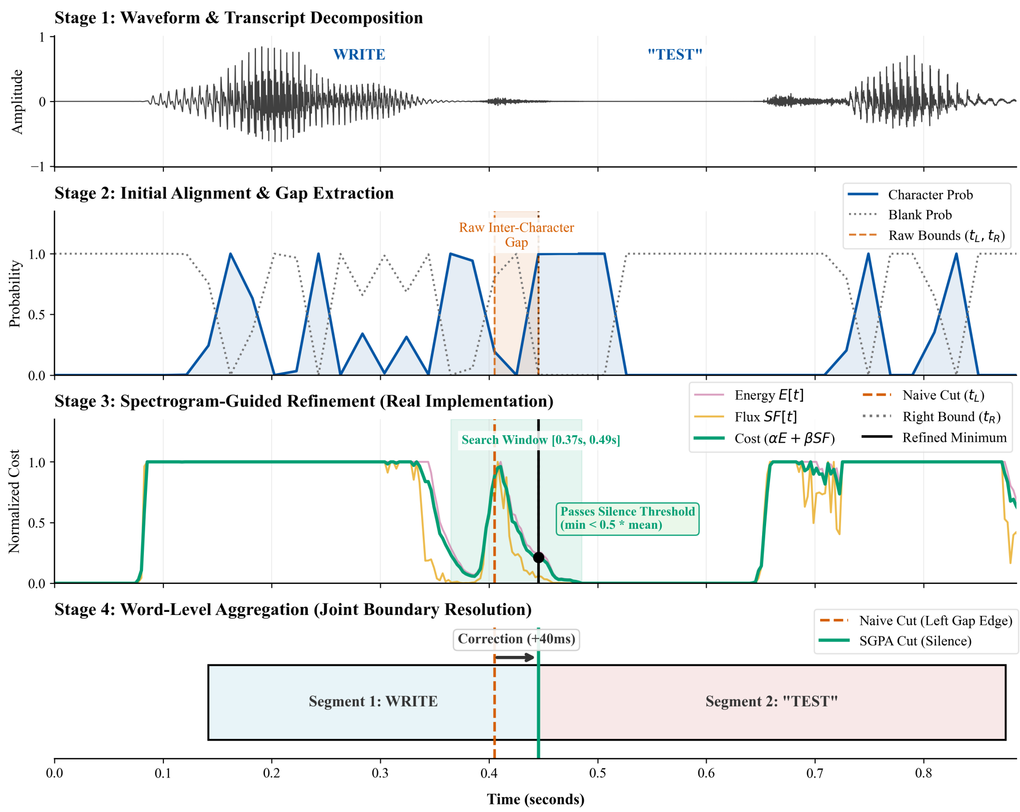

Figure 1: *SGPA pipeline on a sample audio segment from the artificial part of the dataset.* ***(1)*** *The raw audio waveform is decomposed into target transcript segments.* ***(2)*** *Initial alignment identifies a raw inter-character gap bounded by CTC probabilities (t_L, t_R).* ***(3)*** *To prevent phoneme truncation, SGPA defines a localized search window and minimizes a joint spectral cost function (energy $E[t]$ and spectral flux $SF[t]$) to find a spectrally stable minimum.* ***(4)*** *The final word-level boundary is shifted from the naive left-edge cut to the verified silence threshold, resulting in a precise, zero-gap segmentation.*

providing callers with visibility into boundary quality heterogeneity across the player set.

The leading edge of the first character span and the trailing edge of the last character span have no gap neighbour on their outer side; these are refined independently using a fixed ± 80 ms window centred on their respective raw timestamps, subject to the same silence-aware fallback.

2.4. Stage 4: Word-Level Aggregation

The refined character-level boundaries are merged into word-level segments using the membership map from Stage 1. This granularity aligns with the primary unit of human interpretation (words), significantly reducing the player count (Figure 2), and makes SV estimation with the Neyman-allocated complementary-contributions estimator [22, 23] computationally feasible. A word-level segment is marked as boundary-unrefined if any of its constituent character boundaries fell back to the raw gap midpoint, enabling downstream filtering or flagging of lower-quality players.

3. Diagnostic Evaluation

To validate SGPA, we conduct a controlled comparison between two segmentation regimes: (i) the model's native audio tokenization and (ii) SGPA word-aligned segments. All remaining estimator and inference settings are held constant. Reproduction notebooks are available in the project repository[3].

3.1. Setup

Model We use **LFM2-Audio-1.5B** [24][4], a 1.5 B-parameter open-source, end-to-end audio foundation model. Our selection criteria were: (i) direct audio token processing without a cascaded speech $\rightarrow$ text pipeline [1, 2]; (ii) publicly available weights; and (iii) a parameter count small enough to fit within the 16 GB VRAM of a consumer-grade Nvidia RTX 4080. LFM2 was the smallest publicly available model satisfying all three criteria at the time of experimentation.

SV Characteristic Function The characteristic function $v(S)$ is defined as the log-probability assigned by LFM2-Audio-1.5B to the ground-truth target transcription when all players not in coalition S are silenced. Formally, for a transcript y^* and a masked audio signal x_S in which segments outside S are replaced with zero amplitude, $v(S) = \log P(y^* \mid x_S)$.

Datasets We evaluate our approach using two single-sentence English corpora derived from industry-standard benchmarks: (i) `voice_bench_single_sentence_1k`, adapted from **VoiceBench** [25], which features paired

[3] `https://github.com/Pawlo77/MLLM-Shap`
[4] `https://huggingface.co/LiquidAI/LFM2-Audio-1.5B`

Table 1: *Effect of SGPA on Neyman SV estimation cost and phase budget distribution. Mean model calls, total execution time, and proportion of samples per Neyman step count over* 100 *VoiceBench single-sentence prompts.*

				Phase (%)	
SGPA	**Mode**	**Calls (mean)**	**Time (s)**	**1**	**2**
✓	SM2S	59.4	67.5	37	63
✓	SF2S	59.3	66.1	37	63
–	SM2S	2365.8	1696.5	0	100
–	SF2S	2552.1	1819.9	2	98

synthetic (male and female) audio alongside original speech; and (ii) `single_sentence_500`, adapted from **LibriSpeech-ASR** [26], which consists entirely of natural human speech. By leveraging these two datasets, we are able to analyze our contribution comprehensively across both artificial and authentic speech domains. Both curated corpora are publicly released alongside this work[5]. **Due to hardware limitations, we limit our evaluation to the 100 prompts from each dataset.** Full details regarding the construction pipeline—including deduplication, sentence filtering, semantic quality filtering, token-budget filtering, and sampling—are provided in Appendix B.

Modes Since SGPA targets audio inputs, we evaluate two speech-to-speech configurations: male voice (`SM2S`) and female voice (`SF2S`). Each prompt is synthesized in both variants to assess SGPA's consistency across speaker characteristics.

SV approximation We employ the Neyman-allocated complementary-contributions estimator [22], which reuses each coalition's complementary contribution across all players it contains, combined with the Neyman stratified sampling formula [23] for optimal budget allocation. The total budget is set to $3n^2$ model evaluations, where n is the number of players. The algorithm proceeds in two phases: Phase 1 draws $m_{\text{init}} = \max(2, \lfloor m/2n^2 \rfloor)$ samples per stratum to initialize variance estimates; Phase 2 allocates the remaining budget proportionally to the estimated standard deviations, minimizing the total SV variance.

Infrastructure All experiments run on a single Nvidia RTX 4080 GPU (16 GB VRAM). The `mllm-shap` package[6] [14] implements SGPA, the SV estimators, and model connectors.

3.2. Feasibility: Coalition Space Reduction

Table 1 reports computational cost under both segmentation regimes. Without SGPA, native audio tokenization produces long feature sequences and the Neyman estimator requires $\approx$2,552 model calls per sample (mode `SF2S`), with a mean total execution time of 1820 s. With SGPA, word-level aggregation compresses the player set to $\approx$7 segments, reducing calls to $\approx$59 per sample – a 43× reduction – and total execution time to $\approx$66 s (28× faster).

[5] `https://huggingface.co/datasets/Pawlo77/mllm-shap`

[6] `https://pypi.org/project/mllm-shap/`

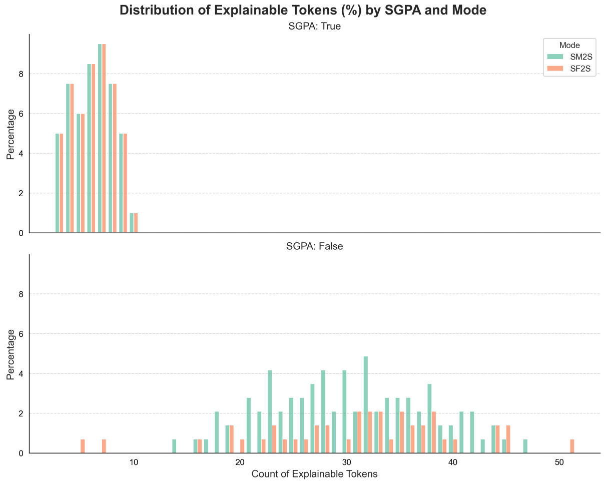

Figure 2: *Distribution of explainable token counts per sample with and without SGPA across modes. SGPA concentrates counts in a narrow regime; native tokenization yields substantially longer and more dispersed sequences.*

Figure 2 illustrates the shift in feature-count distributions. With SGPA enabled, explainable lengths concentrate in a narrow, low-count regime (interquartile range $\approx$2); without SGPA, they expand substantially and become more dispersed, directly increasing coalition complexity.

The Neyman budget allocation also shifts qualitatively (Table 1). With SGPA, the estimator distributes 37% of its budget to the initial variance-estimation phase (Phase 1) and 63% to the optimized phase (Phase 2). Without SGPA, the large game forces nearly all budget ($\geq$ 98%) into Phase 2, leaving minimal room for reliable variance estimation and yielding a less stable approximation.

3.3. Attribution Statistics

Because SGPA redefines the player partition – from model-native audio units to word-aligned segments – it alters the cooperative game being solved. Attribution statistics under the two regimes are therefore expected to differ; the question is how and by how much. To quantify this, we compare three concentration metrics.

3.3.1. Metrics

Let $\tilde{s}_i = |s_i| / \sum_j |s_j|$ denote the normalized absolute SV for player i. We report three complementary concentration metrics. The **Gini coefficient** $G = \frac{1}{2n} \sum_i \sum_j |\tilde{s}_i - \tilde{s}_j|$ is our primary concentration metric: it captures inequality across the full player distribution and requires no length normalization, making it directly comparable across SGPA ($\approx$7 players) and native tokenization ($\approx$50 players).

As secondary characterizations we report **Top-**20% **mass** $\sum_{i=1}^{\lceil 0.2n \rceil} \tilde{s}_{(i)}$ (the share captured by the top quintile) and **normalized entropy** $H_{\text{norm}} = -\sum_i \tilde{s}_i \log \tilde{s}_i \,/\, \sqrt{n}$.[7] These metrics supplement the Gini coefficient with additional characteri-

[7] Raw Shannon entropy grows with n even for a uniform distribution. We divide by $\sqrt{n}$ as a heuristic for partial length normalization when comparing distributions of different sizes; a fully length-invariant alternative is to divide by $\log n$, yielding values in $[0, 1]$. We use $\sqrt{n}$ to avoid over-compressing variance at small player counts, but we note

Table 2: *Paired significance tests of per-observation SV summary metrics (SGPA* on *vs.* off*), Bonferroni-corrected* $\alpha = 0.05$*. Entropy is normalized by* $\sqrt{n}$ *(secondary metric; see text).*

Mode	Metric	p	Cohen's d	Sig.
SM2S	Gini	<0.01	0.50	✓
	Entropy	<0.01	−1.37	✓
	Top-20%	<0.01	0.86	✓
SF2S	Gini	0.01	0.21	—
	Entropy	<0.01	−0.97	✓
	Top-20%	<0.01	0.72	✓

zations of the attribution distribution; our core conclusions do not depend on the entropy normalization choice.

A conceptual note: SGPA does not assert that more concentrated attribution is normatively correct. The shift in attribution statistics observed below is a descriptive consequence of redefining the player partition – an inherent property of Shapley values, which depend on the chosen player set [4]. For a sentence where all words contribute equally, SGPA would assign equal SVs across segments. The entropy and Gini changes reflect the distributional consequences of aggregating ≈50 noisy frame-level attributions into ≈7 interpretable word-level scores.

3.3.2. Results

We apply paired t-tests with Bonferroni correction ($\alpha = 0.05$). Table 2 reports the results. Nearly all comparisons are significant, confirming that SGPA systematically alters attribution concentration.

The Gini coefficient is statistically significant for `SM2S` ($p < 0.01$, $d = 0.50$) and directionally consistent for `SF2S`. This provides length-normalization-free evidence that SGPA shifts attribution concentration. The large negative Cohen's d for normalized entropy (-1.37 for `SM2S`, -0.97 for `SF2S`) indicates that SGPA *increases* entropy, spreading attribution more evenly across the fewer word-level players – consistent with consolidating many noisy frame-level scores into fewer, more stable word-level attributions. The positive d for top-20% mass reflects the mechanical consequence that each player's share grows as the total count shrinks.

The sole non-significant comparison (`SF2S` Gini, d=0.21) suggests that overall inequality is less sensitive to segmentation for female-voice inputs. Figure 3 visualizes this effect: SGPA expands the interquartile range of normalized entropy, indicating a broader set of concentration regimes across observations.

3.4. Preservation of Macro-Trends

A key methodological question is whether SGPA alters the qualitative attribution trajectory. Figure 4 addresses this by comparing position-normalized cumulative SV profiles. Both conditions exhibit an "early mass" pattern – the first few positions accumulate a disproportionate share of total attribution – followed by gradual decay. SGPA modifies local allocation density (the fine-scale derivative structure) while preserving the broader accumulation profile. This is expected: SGPA aggregates fine-grained audio units into semantically interpretable segments, which dampens high-frequency variations attributable to micro-boundary differences under native tokenization, while the global trajectory remains intact.

that our central claims regarding attribution concentration rest on the Gini coefficient, which requires no such normalization.

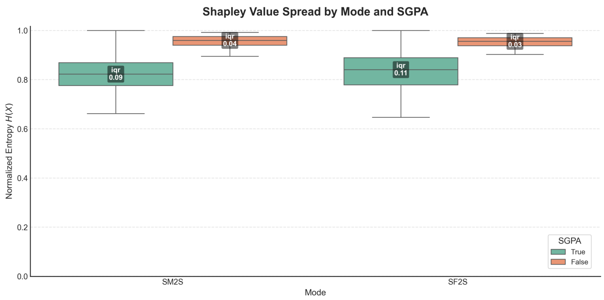

Figure 3: *Normalized attribution entropy ($H/\sqrt{n}$) by mode with and without SGPA. SGPA widens the interquartile range, reflecting greater variability in concentration across samples.*

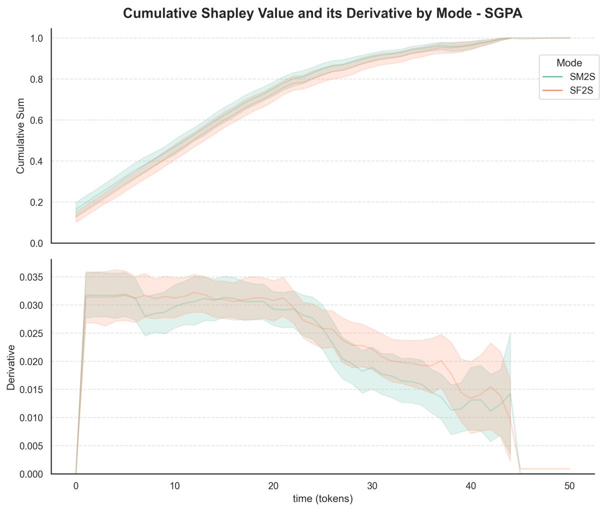

Figure 4: *Position-normalized cumulative SV profile (with derivative) for speech-to-speech mode, illustrating the preserved "early mass" macro-trend under SGPA.*

4. Discussion and Conclusion

We introduced SGPA, a four-stage alignment pipeline that converts raw audio waveforms into word-aligned segments for SV attribution. By combining CTC-based forced alignment with spectrogram-guided boundary refinement, SGPA addresses the three barriers that make SV computation on audio intractable under native tokenization. Our controlled diagnostics on LFM2-Audio-1.5B with VoiceBench yield three principal findings:

1. **Feasibility.** SGPA compresses the player set from ≈50 native tokens to ≈7 word-aligned segments, reducing model evaluations by 43× (from 2,552 to 59) and total execution time from ≈30 minutes to roughly one minute per sample on a consumer-grade GPU.
2. **Non-neutrality.** SGPA fundamentally changes the cooperative game being solved. The Gini coefficient – a length-normalization-free primary metric – is statistically significant for `SM2S` ($d = 0.50$), and directionally consistent for `SF2S`, confirming that word-level segmentation shifts attri-

bution concentration. SGPA also increases normalized entropy ($d = -1.37$ for SM2S) and raises top-quintile mass ($d = 0.86$). This redistribution is an inherent property of Shapley values: explanations depend on the chosen player partition [4], and any change to that partition necessarily changes the resulting attributions. The trade-off is deliberate – interpretability over fidelity to the model's internal tokenization. Crucially, SGPA does not enforce concentrated attribution: for a sentence where all words contribute equally, SGPA assigns equal SVs across segments.

3. **Structural preservation.** Despite altering local allocation density, SGPA preserves the global cumulative SV profile and its characteristic "early mass" pattern, indicating that the word-level partition retains the model's broad attribution structure.

4.1. Limitations

Five limitations should be noted. First, SGPA's spectral refinement weights (α=0.8, β=0.2) were tuned on a limited English TTS set; their robustness to tonal languages or phonologically diverse inventories has not been evaluated. Second, all diagnostics used a single model (LFM2-Audio-1.5B) due to research hardware constraints; replication on larger architectures such as Kimi Audio [27] would strengthen generalizability. Third, because SGPA redefines the player partition, attributions under SGPA and native tokenization are not directly comparable – this is an inherent property of SV, where explanations depend on the chosen player set [4]. Fourth, SGPA requires a transcript; audio-only applications would need upstream ASR, introducing an additional error source. Fifth, the primary evaluation corpus consists entirely of synthesized TTS speech (Google Chirp 3 HD), which exhibits regular inter-word pauses and clean phoneme boundaries. The silence-aware fallback in Stage 3 is designed to degrade gracefully on natural speech – returning the raw gap midpoint when no genuine pause is detected – but the rate of such fallbacks and their effect on SV fidelity under naturalistic conditions remains to be quantified.

4.2. Future work

Future work will extend validation to additional architectures, cross-lingual settings, and alternative SV estimators. Evaluation on natural speech corpora (e.g., LibriSpeech) is a near-term priority, with the `boundary_refined` flag providing a direct per-segment quality signal for analysing the scope and impact of Stage 3 fallbacks. The fixed spectral weighting could be replaced with learned boundary predictors. Phrase-level and sub-word-level aggregation strategies may offer complementary granularity trade-offs for different analytical objectives. A fully length-invariant entropy normalization (division by $\log n$) will be evaluated to complement the Gini-first analysis.

5. Generative AI Use Disclosure

Generative AI tools were used for editing and polishing the manuscript. All content was reviewed and validated by the authors, who take full responsibility.

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A. Hyperparameter Tuning for Spectral Refinement

This appendix reports numeric outputs from `experiments/interspeech/alpha_search.ipynb`, used to choose the Stage 3 boundary-refinement weight α in Eq. 2 ($\beta = 1 - \alpha$).

Search setup. Candidates were $\alpha \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. The automatic pass used 1232 utterances with 3 masks per utterance, yielding 18450 stimuli in total (3690 per α). The manual-validation subset used 20 utterances, i.e., 300 stimuli (60 per α).

Stage 1: automatic proxy score. Lower values are better. On the full automatic set (`auto_all`), mean click scores were: 0.44 ($\alpha = 0.5$), 0.45 ($\alpha = 0.8$), 0.45 ($\alpha = 0.6$), 0.45 ($\alpha = 0.9$), and 0.45 ($\alpha = 0.7$). On the manual subset scored automatically (`manual_auto_20`), ranking changed to: 0.46 ($\alpha = 0.8$), 0.46 ($\alpha = 0.9$), 0.46 ($\alpha = 0.5$), 0.47 ($\alpha = 0.7$), and 0.48 ($\alpha = 0.6$).

Stage 2: human MOS. A single annotator rated all 300 manual-subset stimuli on a 1–5 click MOS scale (1 = severe clicks, 5 = no clicks). For consistency with artifact minimization, lower mean MOS was preferred. Means were: 2.88 ($\alpha = 0.8$), 2.98 ($\alpha = 0.5$), 3.17 ($\alpha = 0.9$), 3.35 ($\alpha = 0.7$), and 3.43 ($\alpha = 0.6$).

Combined decision. We combined automatic and human signals on the manual subset using equal weights after min-max normalization: $\texttt{combined\_score} = 0.5 \cdot \texttt{auto\_norm} + 0.5 \cdot \texttt{human\_norm}$ (lower is better). Final scores were: 0.00 ($\alpha = 0.8$), 0.24 ($\alpha = 0.5$), 0.32 ($\alpha = 0.9$), 0.71 ($\alpha = 0.7$), and 1.00 ($\alpha = 0.6$). We therefore selected $\alpha = 0.8$ and $\beta = 0.2$.

Sensitivity. Mean spectral flux by α was: 0.05 (0.5), 0.05 (0.6), 0.05 (0.7), 0.05 (0.8), and 0.05 (0.9). The range is 0.00 (about 0.56% relative to the minimum), indicating limited sensitivity around the chosen operating point.

B. Dataset Construction Details

We release two single-sentence corpora used in this paper: `voice_bench_single_sentence_1k` and `librispeech_single_sentence_1k`. Both use the same text-first curation pipeline before audio-level processing:

1. **Exact deduplication.** We group by exact prompt/transcript string match and keep one representative row per unique text. Metadata fields (e.g., source dataset labels) are merged; for audio fields we keep the first occurrence.
2. **Sentence filtering.** We segment text with NLTK sentence splitting [28], retain only entries with exactly one sentence, and drop non-English entries using an automatic language classifier.
3. **Semantic quality filtering.** We apply a two-stage filter:
   - *Rule-based pre-filter:* reject entries with fewer than 4 words, more than 300 characters, URL/code-like prefixes, or low alphabetic ratio (< 0.55).
   - *Embedding-based scoring:* encode text with `all-MiniLM-L6-v2`, compute cosine similarity to an anchor centroid, and remove the bottom 20% by score.
4. **Semantic near-duplicate removal.** We run greedy cosine-similarity deduplication on normalized embeddings, ordered by descending quality score, with threshold $\tau = 0.92$.
5. **Token-budget filtering.** We compute explainability token counts from the LFM2 chat mask (with punctuation tokens excluded) and keep only entries with token count ≤ 10.
6. **Sampling.** We perform stratified sampling over token-count strata with deterministic seeding and shortfall backfilling from the remaining pool to reach target size.

The `voice_bench_single_sentence_1k` dataset, derived from the source partitions of VoiceBench, provides three aligned audio variations alongside duration metadata for each prompt: `audio__original`, `audio__male`, and `audio__female`. While `audio__original` preserves the unmodified source recordings, the male and female variants are synthesized via Google Chirp 3 HD TTS[8] using default English (`en-GB`) voice configurations. The final evaluation subset consists of 854 entries restricted to short utterances of at most 10 tokens, as determined by the LFM2 tokenizer on the transcriptions. Sequence lengths are uniformly distributed for entries of 6 to 10 tokens, with slightly reduced frequencies for lengths of 4 to 5 tokens. Overall, the dataset averages 7.61 tokens per entry (SD = 1.70). The mean audio durations are 3.98 s for the original speech, 2.54 s for the synthetic male speech, and 2.74 s for the synthetic female speech.

Similarly, the `single_sentence_500` dataset is sourced from `openslr/librispeech_asr` but retains only the unmodified `audio__original` LibriSpeech recordings, omit-

[8] `https://cloud.google.com/text-to-speech`

ting any TTS synthesis. This dataset also targets short utterances of up to 10 tokens (via the LFM2 tokenizer) and comprises 448 total entries. Its sequence length distribution is right-skewed, with most entries containing 9 or more tokens, yielding a higher mean length of 8.97 tokens (SD $= 1.21$). The mean audio duration for these original recordings is 2.44 s.

For rapid evaluation, we use limited 100-sample versions of both datasets, constructed by selecting the first 20 prompts for each token-count bin from 6 to 10. Reproduction notebooks detailing the data preparation pipeline are available in the repository under the `experiments/data_preparation/` directory.